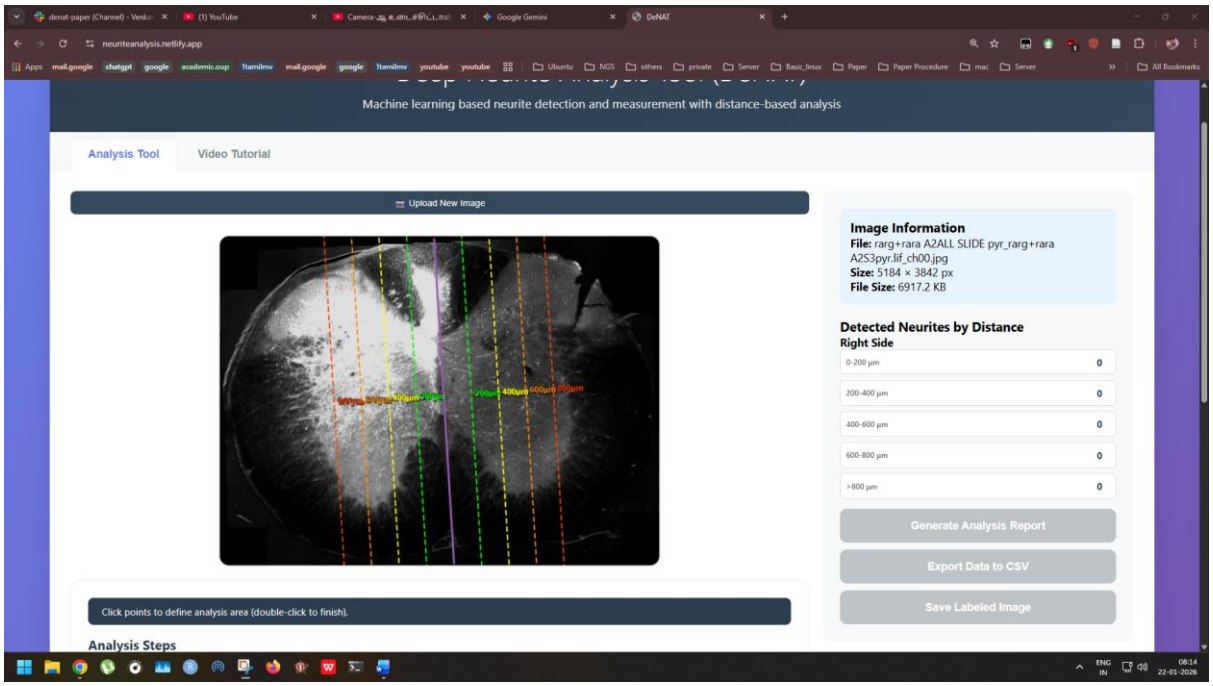
DeNAT Analysis Workflow

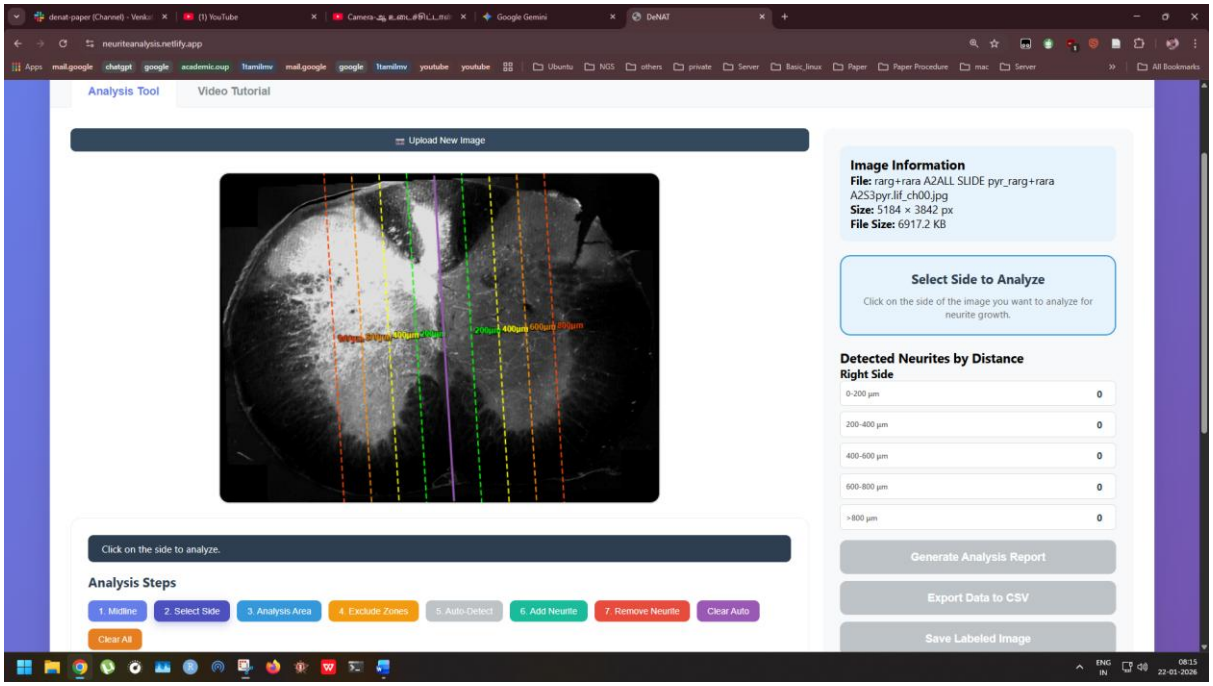
1. Upload and Midline Setup

First, upload your microscopy image. Use the **Midline** tool to draw a vertical reference line. This serves as the "zero" point from which all neurite distances are measured.



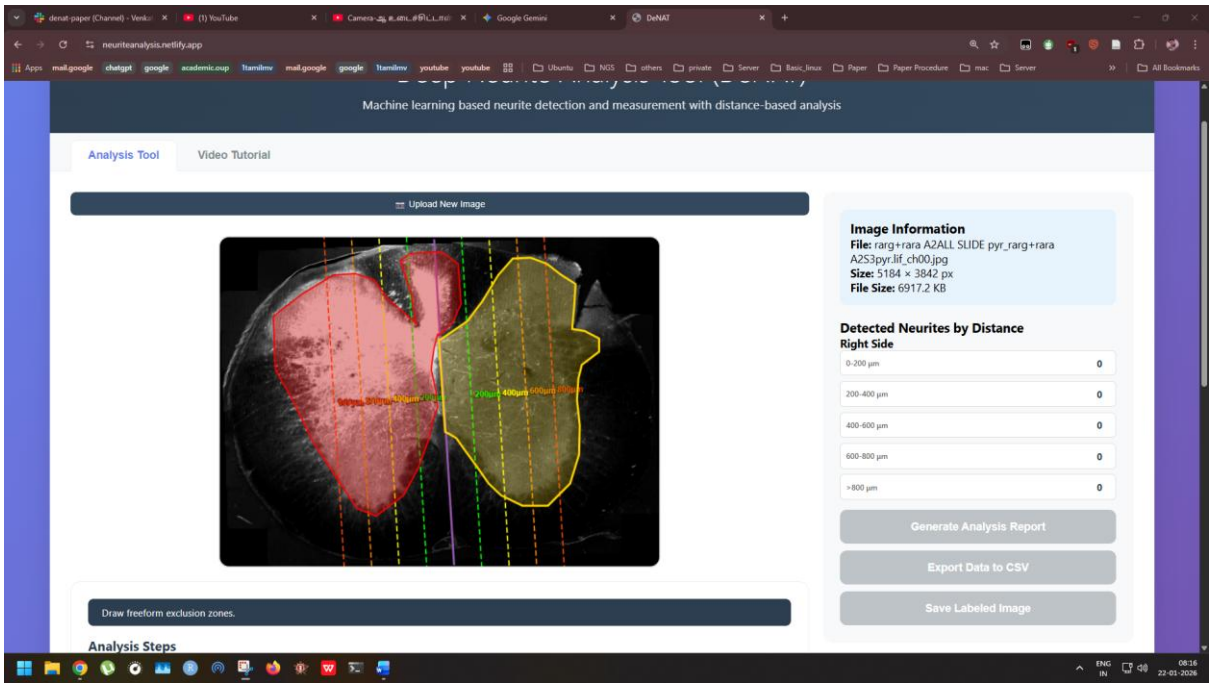
2. Selecting the Analysis Side

Click the **Select Side** button and then click on the side of the midline where you want to measure growth (Left or Right). The tool will highlight this region to confirm the area of interest.



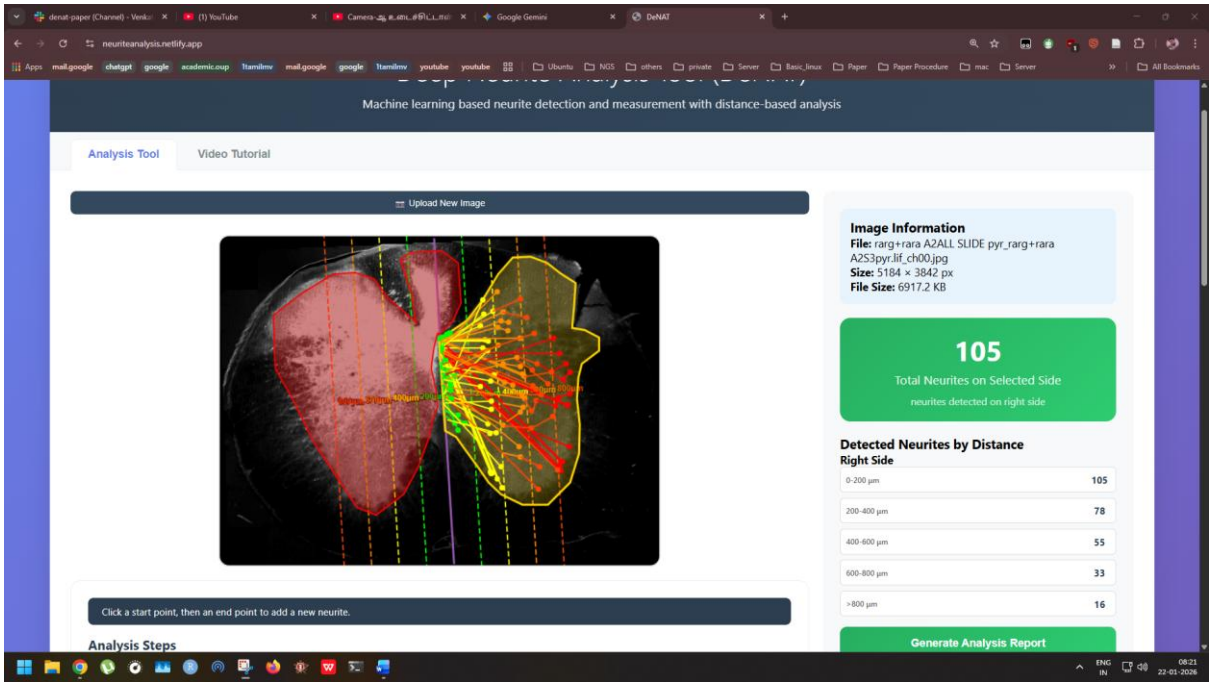
3. Defining the Analysis Area

Use the **Analysis Area** tool to draw a polygon or freeform shape around the specific cluster of neurites you wish to study. This ensures the software only counts relevant signals.



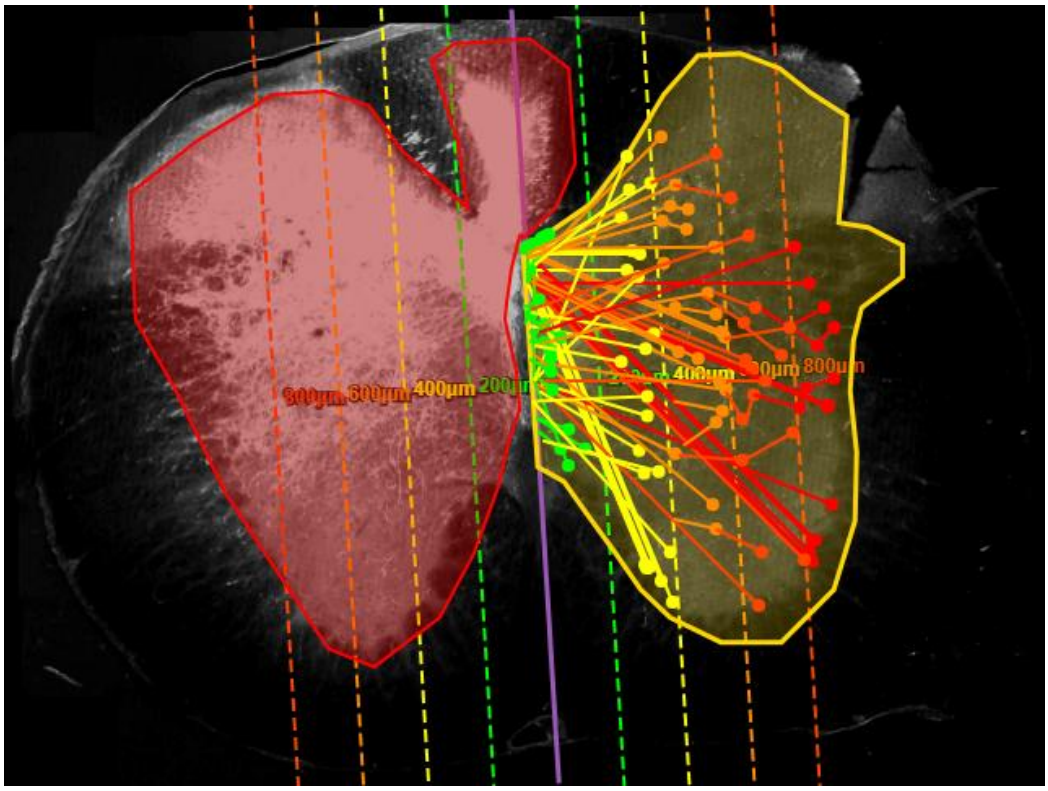
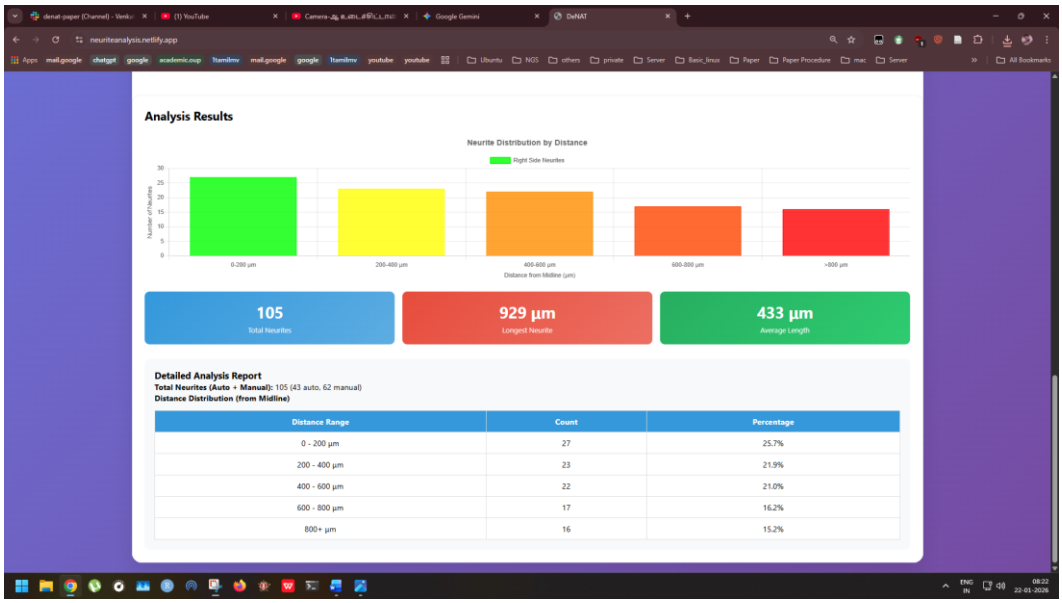
4. Automatic Detection

Click **Auto-Detect**. The machine learning algorithm will identify and trace neurites within your defined area. The results are color-coded based on their distance from the midline (e.g., green for 0–200 µm, yellow for 200–400 µm).



5. Review and Refine (If required)

You can manually add missing neurites using the **Add Neurite** tool or remove false positives with the **Remove Neurite** tool to ensure 100% accuracy before final reporting.

6. Results and Export

The tool generates a comprehensive **Analysis Result** page featuring a histogram of neurite distribution by distance, total neurite count, longest neurite length, and average length. These can be exported as a CSV.